

**Quiz 6: Computer Networks****Total Marks: 20****Group A****CLO 3: Demonstrate operations of various classical routing and switching protocol...****ID:****Name:****Section:**

Total Marks: [2+2+6=10 Marks]

**Q1:** An ISP owns the address block **203.0.113.0/26**. It wants to create **five equal-sized subnets**. Write the prefix (a.b.c.d/x) for each subnet. [2]

Solution: **Subnets:**

1. **203.0.113.0/29**
2. **203.0.113.8/29**
3. **203.0.113.16/29**
4. **203.0.113.24/29**
5. **203.0.113.32/29**

**Q2:** Suppose datagrams are limited to 1,500 bytes (including header) between source Host A and destination Host B. Assuming a 20-byte IP header, how many datagrams would be required to send an MP3 consisting of 5 million bytes? Explain how you computed your answer. [2]

$$1500 - 20 = \mathbf{1480 \text{ bytes}}$$

$$5,000,000 / 1480 \approx 3378.38 \Rightarrow 3379 \text{ datagrams}$$

**Answer: 3379 datagrams.**

**Q3:** Three flows share a link with weights 3, 1, and 2 respectively. Their packet sizes are 150 bytes, 450 bytes, and 300 bytes.

i. Calculate the total number of bytes transmitted by all flows in one complete WFQ round, assuming each flow transmits data proportional to its weight. [2]

We want bytes in ratio **3 : 1 : 2**, so find integer packets per flow:

$$n_1 \cdot 150 : n_2 \cdot 450 : n_3 \cdot 300 = 3 : 1 : 2$$

Solution (smallest integers):

- $n_2 = 1$
- $n_1 = 9$
- $n_3 = 3$

So bytes per WFQ “round”:

- Flow 1:  $9 \times 150 = \mathbf{1350 \text{ bytes}}$
- Flow 2:  $1 \times 450 = \mathbf{450 \text{ bytes}}$
- Flow 3:  $3 \times 300 = \mathbf{900 \text{ bytes}}$

Ratios:  $1350 : 450 : 900 = \mathbf{3 : 1 : 2}$

**Total bytes in one WFQ round =  $1350 + 450 + 900 = 2700 \text{ bytes}$**

ii. In a Round Robin scheduler that serves one packet per flow per cycle, what is the total number of bytes transmitted per cycle and what is the order of packet transmission? [2]

One packet each:  $150 + 450 + 300 = \mathbf{900 \text{ bytes per cycle}}$

Order: **Flow 1  $\rightarrow$  Flow 2  $\rightarrow$  Flow 3**

iii. Which scheduler (WFQ or Round Robin) provides better fairness in terms of weighted bandwidth allocation? Justify briefly. [2]

**WFQ** is fairer, because its byte allocation matches the **weights 3:1:2**, while Round Robin gives bytes in ratio  **$150:450:300 = 1:3:2$** , which does **not** respect the configured weights.